Activity

Visit a nearby health center to observe the analysis of urine. Dip strips can be used to test urine for a range of different factors such as pH, glucose, ketones and proteins. Dip sticks used for detecting glucose contain two enzymes namely, glucose oxidase and peroxidase. These two enzymes are immobilized on a small pad at one end of the stick. The pad is immersed in urine. If the urine contains glucose, a brown coloured compound is produced. The resulting colour pad is matched against a colour chart. The colour does not indicate the current blood glucose concentrations.

CASE STUDY

Both the kidneys of Ravi (28 years) were not functioning and he was undergoing dialysis. He was admitted to a hospital with renal failure. His mother Suganthi (47 years) was willing to donate one of her kidneys to her son after she was given counseling. Their blood groups were matching and later approval was obtained from transplant committee and technical committee. Operation was performed for 5 hrs. He was administered with immunosuppressive drugs and anti inflammatory drugs. He recovered from the operation and returned home.

1. Name the disease Ravi was suffering from.
2. What relation is the donor of the kidney
3. Name the type of matching done to perform the transplant.
4. Why approval has to be got from transplant committee and technical committee?
5. What do you think about Suganthi donating her kidney?

Evaluation

1. Concentration of urine depends upon which part of the nephron
   a. Bowman's capsule
   b. Length of Henle's loop
   c. P.C.T.
   d. Network of capillaries arising from glomerulus

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2. If Henle's loop were absent from mammalian nephron, which one of the following is to be expected?
   a. There will be no urine formation
   b. There will be hardly any change in the quality and quantity of urine formed
   c. The urine will be more concentrated
   d. The urine will be more dilute
3. What will happen if the stretch receptors of the urinary bladder wall are totally removed?
   a. Micturition will continue
   b. Urine will be continue to collect normally in the bladder
   c. There will be micturition
   d. Urine will not collection the bladder
4. The end product of Ornithine cycle is
   a. carbon dioxide b. uric acid
   c. urea d. ammonia

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